ICT Engineering for Smart Societies

Course: Advanced Machine Learning for Image Processing

Laboratory 3 Report

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1 An Image Search Engine using CLIP

Abstract

This report describes the implementation of an image search engine using CLIP (Contrastive Language-Image Pretraining), a deep learning model for aligning images and text. The main objective is to develop an efficient system for searching images based on textual queries or reference images, leveraging cosine similarity between feature vectors extracted from images. The work includes loading and preprocessing the Flickr8k dataset, extracting image features using the CLIP ViT-B/32 encoder, and implementing a search function that returns the most similar images. The results demonstrate the system's ability to retrieve relevant images, highlighting CLIP's effectiveness for multimodal search tasks, with potential applications in computer vision and information retrieval.

2 Introduction

2.1 CLIP Architecture

We utilized the **ViT-B/32** model, which employs a Vision Transformer as the image encoder and a Transformer for text processing. CLIP is trained to project both images and text into a shared latent space where semantically related concepts are positioned closely together.

2.2 Database Creation

To build the search engine index:

1. We loaded the Flickr8k dataset.
2. Each image was passed through the CLIP `preprocess` function to resize and normalize it.
3. We used the image encoder to extract a feature vector of size $F = 512$ for each image.
4. All vectors were stored in a "database" matrix of size (N, F), where N is the number of images.

2.3 Search Mechanism

The similarity between a query (text or image) and the database entries is calculated using the **Cosine Similarity**:

$$\text{similarity} = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\| \|\mathbf{d}_i\|} \tag{1}$$

Since CLIP typically returns normalized features, this simplifies to a simple dot product.

3 Experimental Results

3.1 Text-to-Image Search: "A boat"

We encoded the string "A boat" using the CLIP text encoder and retrieved the top 5 images with the highest similarity score.

(a) CosSim: 0.2876

(b) CosSim: 0.2758

(c) CosSim: 0.2732

(d) CosSim: 0.2726

(e) CosSim: 0.2719

Figure 1: 5 images with the highest cosine similarity

3.2 Image-to-Image Search: `1786425974_c7c5ad6aa1.jpg`

We used a reference image from the dataset as a query to find the most similar images in the collection.

(a) CosSim: 1.0000

(b) CosSim: 0.8597

(c) CosSim: 0.8532

(d) CosSim: 0.8502

(e) CosSim: 0.8466

Figure 2: Top 5 retrieved images for the reference image query.

4 Memory Occupation Analysis

The memory required to store the image features depends on the number of images (N) and the dimensionality of the features (F).

- **Number of images (N):** 8,091 (Flickr8k).
- **Feature size (F):** 512.
- **Data precision:** Using `float16` (2 bytes per element).

The total memory occupation is calculated as:

$$\text{Memory} = N \times F \times \text{size\_of(float16)} = 8,091 \times 512 \times 2 \text{ bytes}$$

$$\text{Memory} = 8,285,184 \text{ bytes} \approx \mathbf{8.285} \text{ MB}$$

5 Conclusion

The CLIP model proves to be highly effective for zero-shot image retrieval. By mapping images and text into the same vector space, we built a functional search engine that understands semantic content without requiring specific training on the Flickr8k labels.